Introduction to Economics

Class 10 — Market failures, externalities and public goods

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Universitat de Barcelona · Group A6

26 October 2026

Quiz 7

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle and circle the new letter.

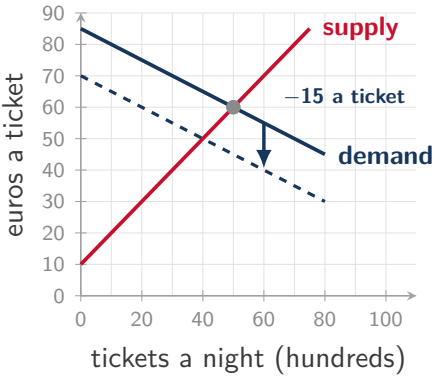
Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Quiz 7
10:11	Warm-up
10:16	Since 2005: a permit for every tonne of CO_2
10:19	When a deal hits a bystander
10:22	Too much glass: the lost triangle, and a tax that fixes it
10:34	In pairs: vans in the old town
10:40	Too few flu jabs; the noisy terrace
10:50	Break
11:00	Rules, taxes or permits?
11:12	The EU's permits: what happened
11:15	In pairs: two power plants
11:21	Four kinds of goods; public goods
11:30	Fish in a bay; in pairs: lights for a courtyard
11:40	In one line; Problem set 2; before Thursday 29 October

Warm-up: a tax paid by the buyers — who pays what?

Flamenco shows: a tax of 15 a ticket

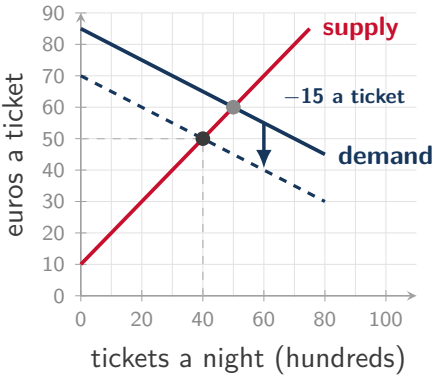


1. Tickets now, and what venues keep?
2. What does a buyer pay in all?
3. Who bears more of the 15, and why?

In pairs, two minutes; then hands per item.

Warm-up: a tax paid by the buyers — who pays what?

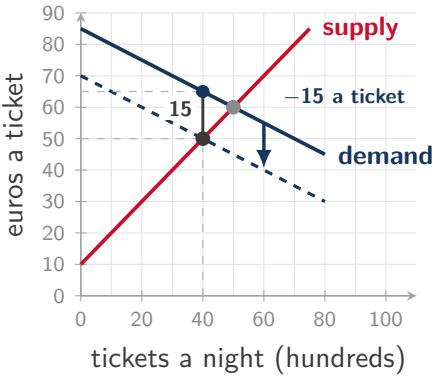
Flamenco shows: a tax of 15 a ticket



1. Tickets now, and what venues keep?
   4,000 a night at 50: the new crossing
2. What does a buyer pay in all?
3. Who bears more of the 15, and why?

Warm-up: a tax paid by the buyers — who pays what?

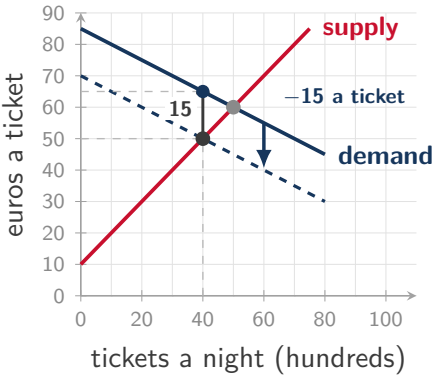
Flamenco shows: a tax of 15 a ticket



1. Tickets now, and what venues keep?
   4,000 a night at 50: the new crossing
2. What does a buyer pay in all?
   50 + 15 = 65: on the solid demand
3. Who bears more of the 15, and why?

Warm-up: a tax paid by the buyers — who pays what?

Flamenco shows: a tax of 15 a ticket



1. Tickets now, and what venues keep?
   4,000 a night at 50: the new crossing
2. What does a buyer pay in all?
   50 + 15 = 65: on the solid demand
3. Who bears more of the 15, and why?
   venues: 10 less; buyers: 5 more
   venues cannot walk away: less elastic

Warm-up: consumer surplus, buyer by buyer

Four friends, one tandem paragliding flight each at most

The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?

2. Their consumer surplus?

3. Does the friend who would pay 70 count?

In pairs, one minute; then hands per item.

Warm-up: consumer surplus, buyer by buyer

Four friends, one tandem paragliding flight each at most

The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?
   three: those who would pay 150, 125 and 100
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Warm-up: consumer surplus, buyer by buyer

Four friends, one tandem paragliding flight each at most

The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?
   three: those who would pay 150, 125 and 100
2. Their consumer surplus?
   $60 + 35 + 10 = 105$ euros
3. Does the friend who would pay 70 count?

Warm-up: consumer surplus, buyer by buyer

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The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?
   three: those who would pay 150, 125 and 100
2. Their consumer surplus?
   $60 + 35 + 10 = 105$ euros
3. Does the friend who would pay 70 count?
   no: does not buy at 90, so adds 0, not -20

Since 2005: one permit for every tonne of CO_2

a power station or a steelworks in the EU → **hands in one permit for every tonne of CO_2 it emits**

The EU limits the number of permits; firms trade them.
In 2024 one cost about 65 euros.

Did emissions fall — and by how much?

EU Emissions Trading System (ETS).

European Commission, Carbon market report COM(2025) 735, 3 December 2025.

An externality: a deal between two that hits a third

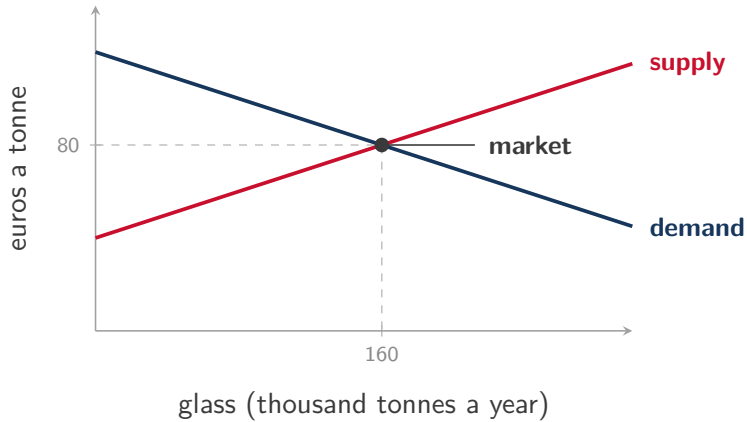


Negative: a bar's terrace keeps the street awake

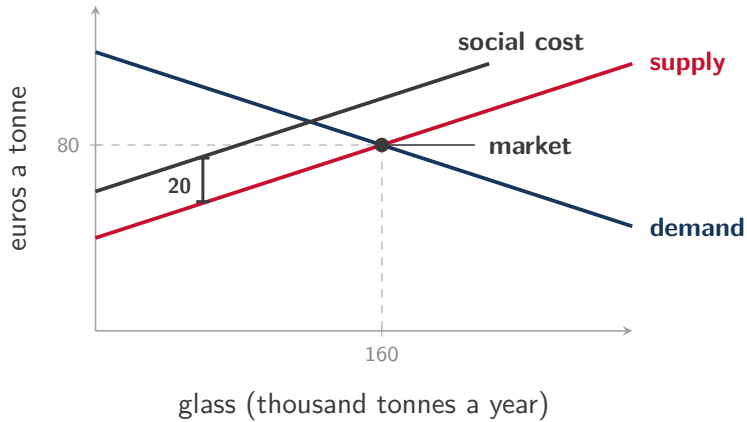
Positive: your flu jab protects your flatmates

The price counts the deal, not the bystander.

Glass: the market makes 160; counting the smoke, 120 is best

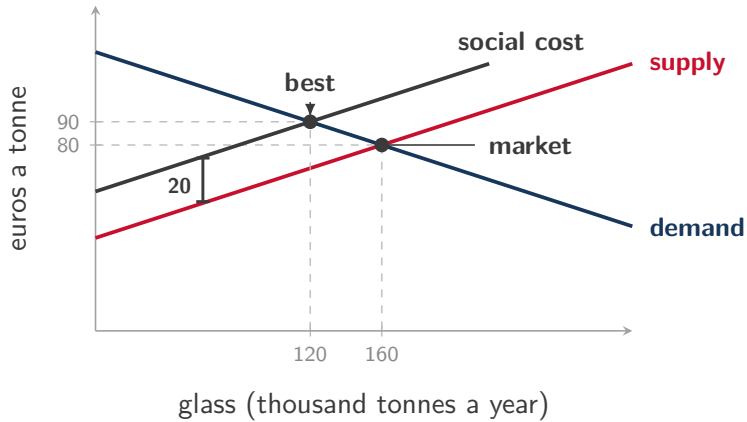


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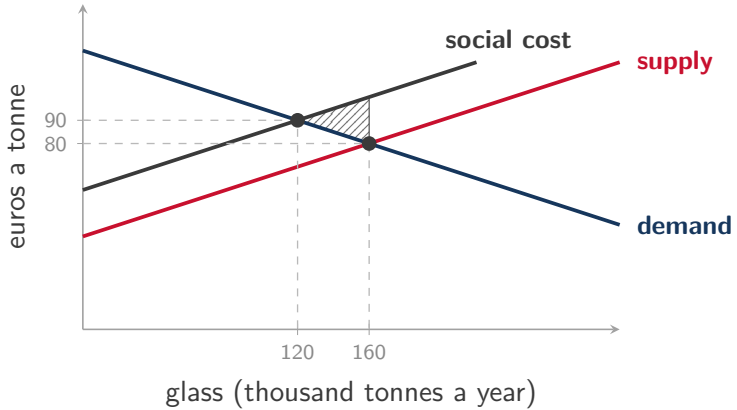
Social cost = the works' own cost + 20 euros of smoke a tonne.

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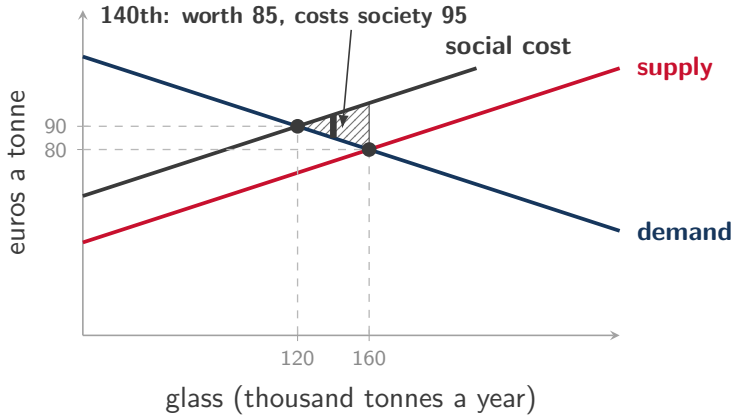


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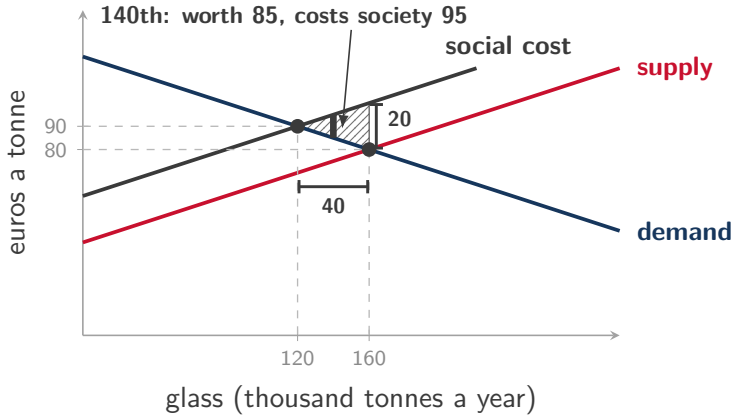
Tonnes 120 to 160 cost society more than they are worth: 400 thousand euros a year



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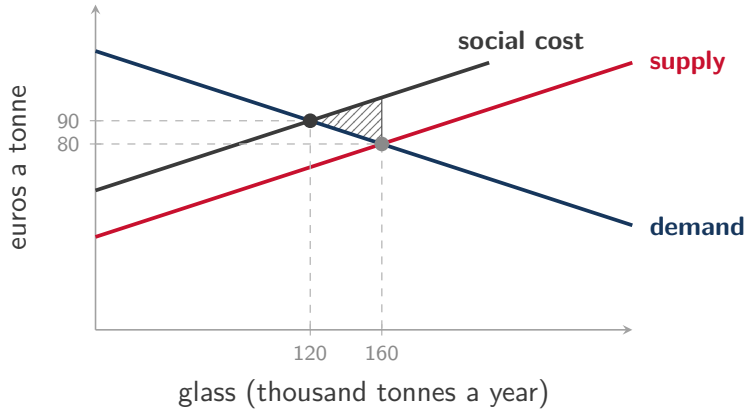


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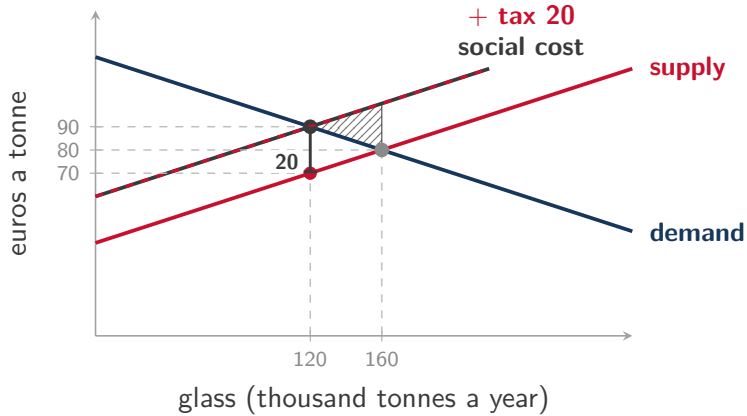


Lost: ½ × 20 × 40 = 400 thousand euros a year — Class 9's triangle, now from too much.

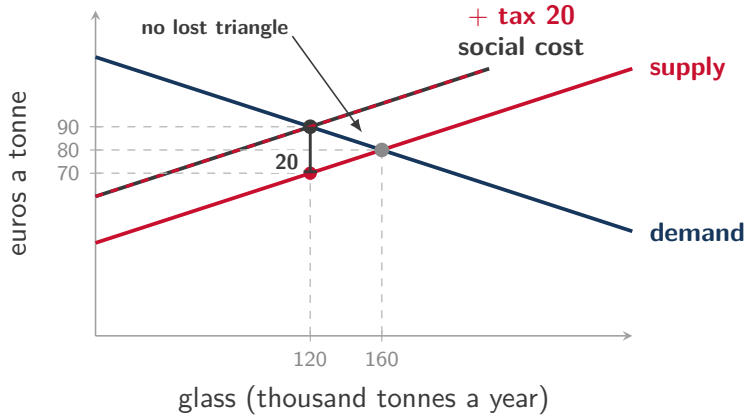
A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone



A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone



A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone



Pigovian tax: a tax that corrects a negative externality; set at the external cost (20).

In pairs, four minutes: vans in the old town

Deliveries in the old town

deliveries a day	last one is worth	last one costs the firm
160	22	14
200	20	16
240	18	18

Each delivery's fumes and noise cost residents 4 euros.

1. How many deliveries does the market make?
2. How many are best, counting the residents?
3. Which tax gets the market there?
4. The lost surplus from making 240, not 200?

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   240: worth 18 = cost 18
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   4 euros a delivery
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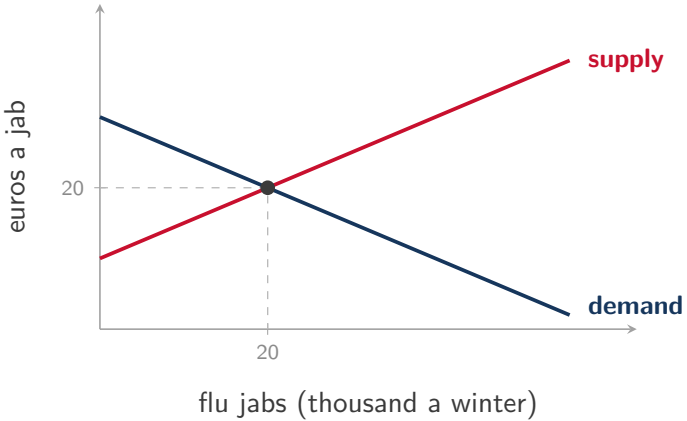
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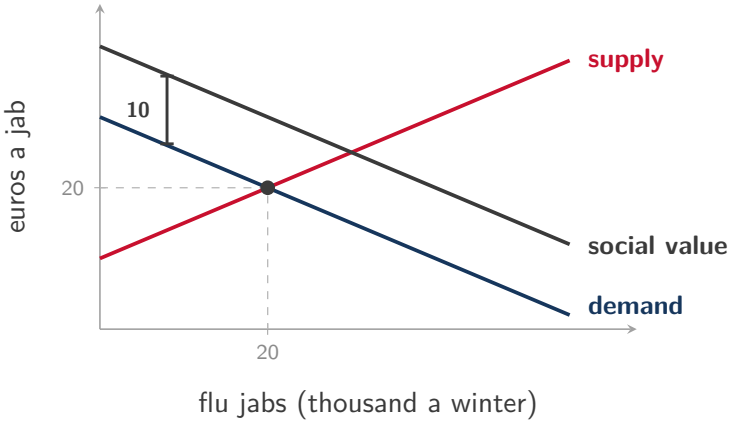
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2. How many are best, counting the residents?
   200: worth 20 = 16 + 4
3. Which tax gets the market there?
   4 euros a delivery
4. The lost surplus from making 240, not 200?
   ½ × 4 × 40 = 80 euros a day

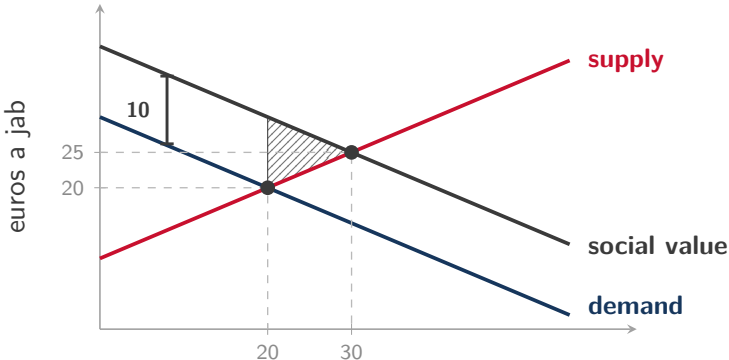
Flu jabs protect others too: the market gives 20 thousand, society wants 30



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Flu jabs protect others too: the market gives 20 thousand, society wants 30



Too few: a subsidy of 10 a jab gets the market to 30.

The noisy terrace: does it matter who holds the right?

A bar's terrace, open late

The bar gains 200 a night · the noise costs the 30 neighbours 300 a night

1. The bar has the right to stay open. What happens?

2. The neighbours have the right to quiet. Now?

3. The bar gains 400 a night instead: open or closed?

In pairs, two minutes; then hands.

The noisy terrace: does it matter who holds the right?

A bar's terrace, open late

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1. The bar has the right to stay open. What happens?
   the neighbours pay it 200 to 300 to close
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   it closes: the bar would offer at most 200
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   it closes: the bar would offer at most 200
3. The bar gains 400 a night instead: open or closed?
   open either way: the right decides who pays

A tax on plastic cups: which curve moves? (optional)

A tax of 0.10 on every plastic cup, paid by the makers

1. Which curve shifts, which way?
2. Cups get dearer: does demand shift too?
3. Reusable cups get cheaper. Now?

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   supply of cups: up by 0.10
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   supply of cups: up by 0.10
2. Cups get dearer: does demand shift too?
   no: fewer cups bought, along demand
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   no: fewer cups bought, along demand
3. Reusable cups get cheaper. Now?
   demand for plastic cups shifts left

Break

Back at 11:00

Thirty neighbours rarely bargain: so the council sets a rule or a price

- Thirty neighbours: who organises them, who pays the lawyer?
- Each hopes the others will pay
- Each side holds out for a better deal
- So the council steps in: a closing hour, or a charge

Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

	Works A	Works B	cost of cutting (thousand euros)
CO_2 now (thousand tonnes a year)	4	4	
cutting a tonne costs (euros)	10	40	

Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

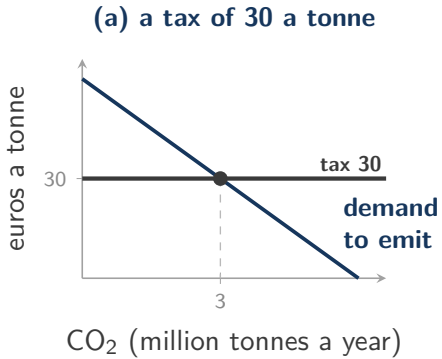
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Rule: each cuts 2	2×10	2×40	**100**

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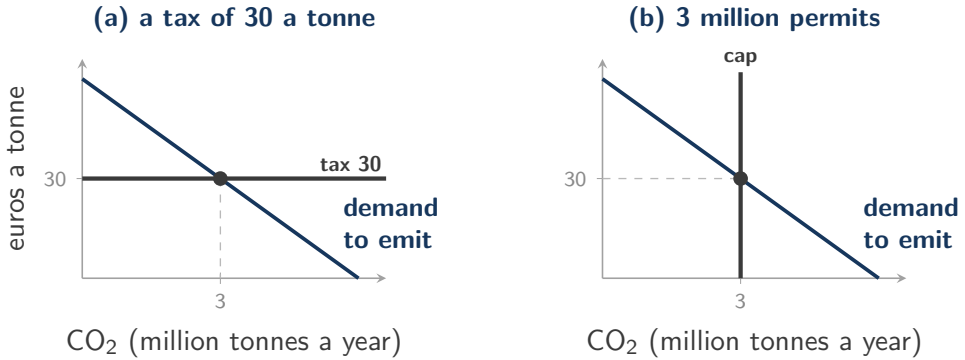
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Rule: each cuts 2	2×10	2×40	**100**
Tax 25 a tonne emitted	cuts 4	cuts none	$\mathbf{4 \times 10 = 40}$

The cheap cutter does the cutting.

A tax fixes the price of emitting; a permit cap fixes the quantity

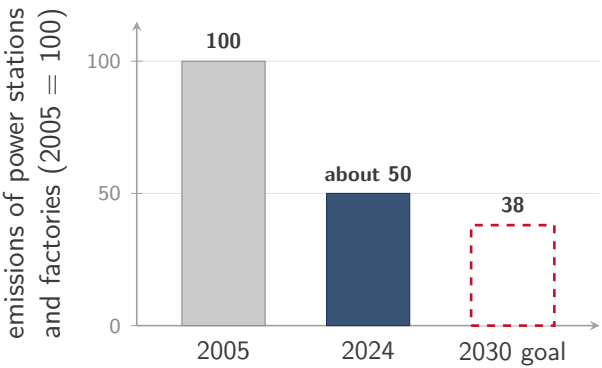


A tax fixes the price of emitting; a permit cap fixes the quantity



Same point: a tax picks the price, a cap picks the quantity.

EU permits since 2005: power stations and factories emit about half as much



The cap shrinks every year; the price decides who cuts.

European Commission: COM(2025) 735, 3 December 2025; verified EU ETS data for 2025, 10 April 2026.

In pairs, four minutes: two power plants, a rule, a tax, permits

	Plant N	Plant S
CO_2 now (thousand tonnes a year)	6	6
cutting a tonne costs (euros)	20	50

Goal: 6 thousand tonnes less in all.
Costs in thousand euros.

1. Rule: each cuts 3. Cost?
2. Tax 30 a tonne emitted: who cuts?
3. The cost of the cut with the tax?
4. Tradable permits to emit 3 each: who sells?

In pairs, four minutes: two power plants, a rule, a tax, permits

	Plant N	Plant S
CO_2 now (thousand tonnes a year)	6	6
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Goal: 6 thousand tonnes less in all.
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1. Rule: each cuts 3. Cost?
   $3 \times 20 + 3 \times 50 = 210$ thousand euros
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   N cuts all 6; S cuts none
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2. Tax 30 a tonne emitted: who cuts?
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3. The cost of the cut with the tax?
   $6 \times 20 = 120$ thousand euros
4. Tradable permits to emit 3 each: who sells?

In pairs, four minutes: two power plants, a rule, a tax, permits

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   $6 \times 20 = 120$ thousand euros
4. Tradable permits to emit 3 each: who sells?
   N sells 3 to S, at 20 to 50 a tonne

Two questions sort goods into four kinds

	excludable (non-payers kept out)	**not excludable** (anyone can use it)
rival (my use leaves less for you)		
not rival (it does not)		

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	excludable (non-payers kept out)	**not excludable** (anyone can use it)
rival (my use leaves less for you)	**private good** a cup of coffee	
not rival (it does not)		

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not rival (it does not)		

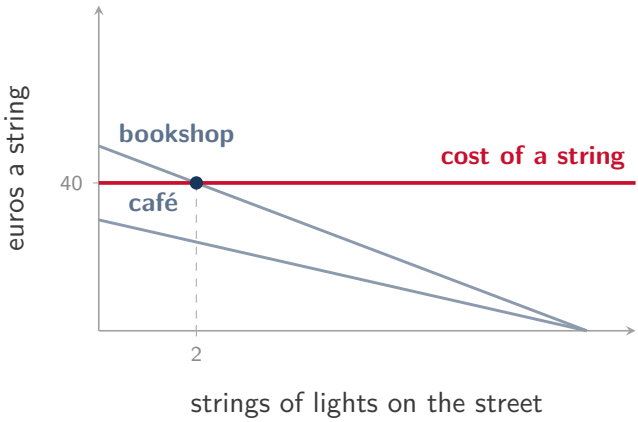
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not rival (it does not)	**club good** a toll motorway, no jams	

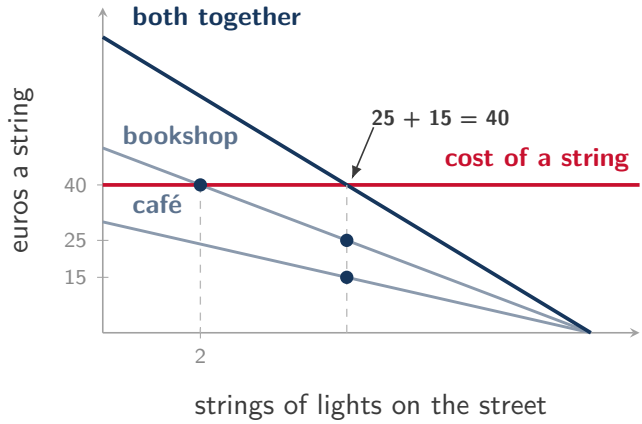
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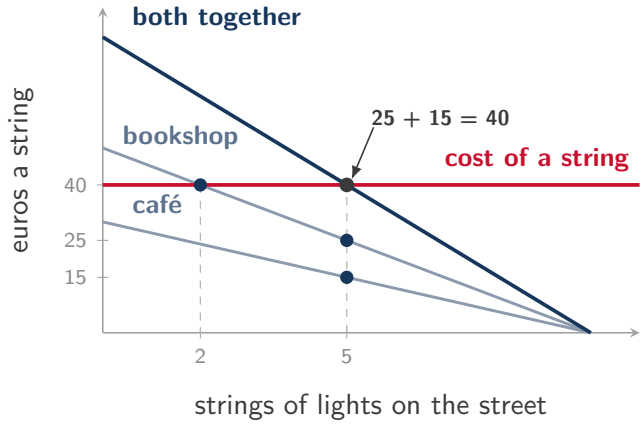
A public good: add what each would pay for each string — the street wants 5



A public good: add what each would pay for each string — the street wants 5



A public good: add what each would pay for each string — the street wants 5



Both get the same lights: add what each would pay.

A free bay fills with boats: each new boat takes fish from the others

Boats in a bay

boats	catch in all	per boat
	(thousand euros of fish a week)	
1	12	12
2	22	11
3	30	10
4	36	9
5	40	8
6	42	7

A boat's week costs 7.

A free bay fills with boats: each new boat takes fish from the others

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4	36	9
5	40	8
6	42	7

A boat's week costs 7.

4th boat: catches 9, but the bay gains only 6

A free bay fills with boats: each new boat takes fish from the others

Boats in a bay

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	(thousand euros of fish a week)	
1	12	12
2	22	11
3	30	10
4	36	9
5	40	8
6	42	7

A boat's week costs 7.

4th boat: catches 9, but the bay gains only 6

Open bay: six boats, each just covers its 7

A free bay fills with boats: each new boat takes fish from the others

Boats in a bay

boats	catch in all	per boat
	(thousand euros of fish a week)	
1	12	12
2	22	11
3	30	10
4	36	9
5	40	8
6	42	7

Rival, open to all: used too much.

A boat's week costs 7.

4th boat: catches 9, but the bay gains only 6

Open bay: six boats, each just covers its 7

Best: three boats; $30 - 21 = 9$ a week left

In pairs, three minutes: lights for a shared courtyard

Two flats share a courtyard; a string costs 25

	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
Flat B would pay	20	15	10	5

1. Flat A alone: how many strings?
2. Flat B alone?
3. Both together?
4. Why might they end up with fewer?

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	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
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1. Flat A alone: how many strings?
   1: the 2nd is worth 20, less than 25
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   none: 20 is less than 25
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In pairs, three minutes: lights for a shared courtyard

Two flats share a courtyard; a string costs 25

	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
Flat B would pay	20	15	10	5

1. Flat A alone: how many strings?
   1: the 2nd is worth 20, less than 25
2. Flat B alone?
   none: 20 is less than 25
3. Both together?
   2: the 2nd is worth 35; the 3rd only 20
4. Why might they end up with fewer?

In pairs, three minutes: lights for a shared courtyard

Two flats share a courtyard; a string costs 25

	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
Flat B would pay	20	15	10	5

1. Flat A alone: how many strings?
   1: the 2nd is worth 20, less than 25
2. Flat B alone?
   none: 20 is less than 25
3. Both together?
   2: the 2nd is worth 35; the 3rd only 20
4. Why might they end up with fewer?
   each counts only its own value: B free-rides

What does a morning's beach clean-up cost you? (optional)

Saturday: 4 hours cleaning the beach

You could earn 11 an hour instead · the bus to the beach costs 3
Your monthly gym fee is already paid

1. Your opportunity cost of the morning?

2. Is the gym fee part of it?

3. Why do too few people volunteer?

What does a morning's beach clean-up cost you? (optional)

Saturday: 4 hours cleaning the beach

You could earn 11 an hour instead · the bus to the beach costs 3
Your monthly gym fee is already paid

1. Your opportunity cost of the morning?
   $4 \times 11 + 3 = 47$ euros
2. Is the gym fee part of it?
3. Why do too few people volunteer?

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1. Your opportunity cost of the morning?
   $4 \times 11 + 3 = 47$ euros
2. Is the gym fee part of it?
   no: paid whatever you do (sunk)
3. Why do too few people volunteer?
   the clean beach is for everyone: a public good

A tax on petrol: positive or normative? (optional)

A tax on petrol: three statements

1. "A tax of 0.20 a litre cuts driving more than one of 0.10."

2. "Cleaner air is worth dearer driving."

3. Both want cleaner air. B: "Car-free zones instead: drivers barely react to prices." B's reason: values or a prediction?

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3. Both want cleaner air. B: "Car-free zones instead: drivers barely react to prices."
   B's reason: values or a prediction?
   a prediction: how drivers respond

A cost on a bystander:
the market makes too much;
a tax equal to that cost fixes it.

Open to all: a public good gets too little,
a common resource is used too much.

A price lets the cheapest cutter do the cutting.
Thursday: what it costs a firm to make things.

Before Thursday 29 October

- Thursday: production and costs
- Read Mankiw & Taylor: Ch. 5, pp. 105–122
- Problem set 2 (Block 3): handed out today; due Monday 2 November, on paper, in class
- Practise in Sextant: the units the course page lists

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